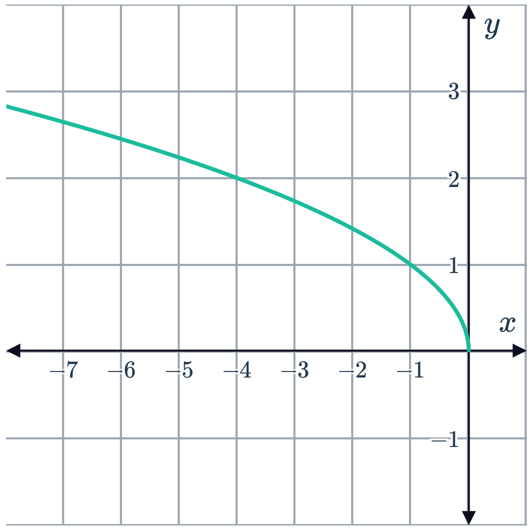
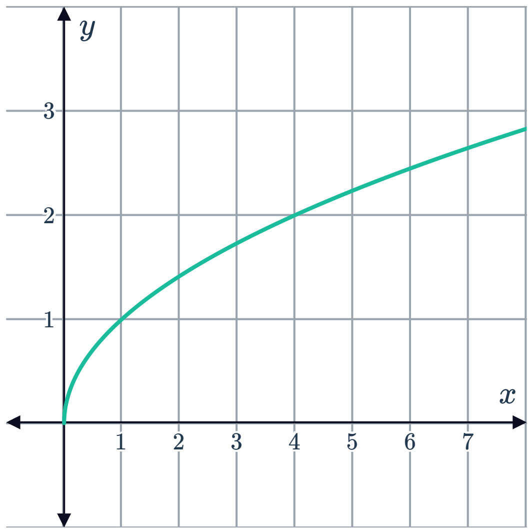
Key features

1 Consider the graph of $y = \sqrt{-x}$:
As x approaches $-\infty$, what does y approach?



2 Consider the graph of $y = \sqrt{x}$:



- **a** Describe the rate of increase of the function as x increases.
- **b** State the axes intercepts.
- **c** Does the function have an asymptote?
- **d** Does the function have a limiting value?
- **e** As x increases, what does y approach?

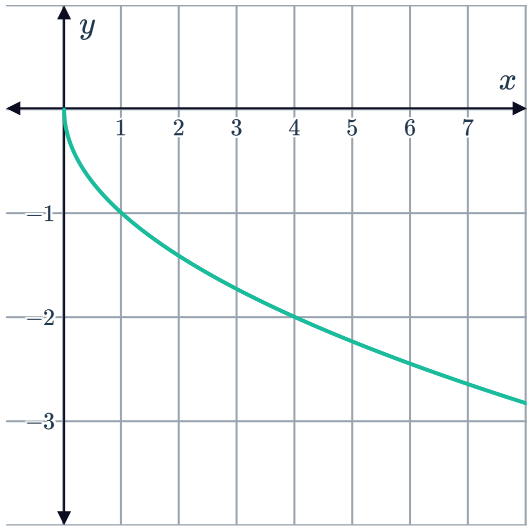
3 Consider the function $y = -\sqrt{x}$.

a Complete the table of values. Round any values to two decimal places if necessary.

x	0	1	2	3	4	5	9
y							

b Can the function values ever be positive?

c The graph of the function $y = -\sqrt{x}$ is shown. Is $y = -\sqrt{x}$ an increasing function or a decreasing function?



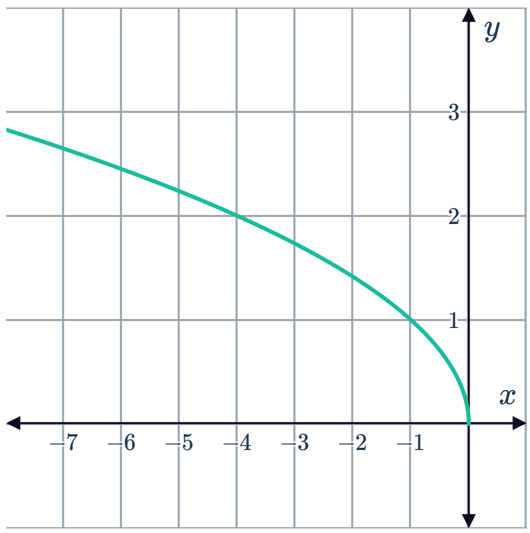
d Describe the rate of decrease of the function as x increases.

4 Consider the function $y = \sqrt{-x}$.

a Complete the table of values. Round any values to two decimal places if necessary.

x	-5	-4	-3	-2	-1	0
y						

b The graph of $y = \sqrt{-x}$ is given. Is $y = \sqrt{-x}$ an increasing function or a decreasing function?



5 Consider the function $y = \sqrt{x} + 3$.

a Can y ever be negative?

b As x gets larger and larger, what value does y approach?

c Determine the y-intercept of the curve.

d How many x-intercepts does it have?

e Sketch the graph.

6 Consider the function $y = 2\sqrt{x} + 3$.

a Is the function increasing or decreasing from left to right?

b Is the function more or less steep than $y = \sqrt{x}$?

c What are the coordinates of the vertex?

d Sketch the graph.

7 Consider the function $y = -\frac{1}{2}\sqrt{x} + 2$.

a Is the function increasing or decreasing from left to right?

b Is the function more or less steep than $y = \sqrt{x}$?

c What are the coordinates of the vertex?

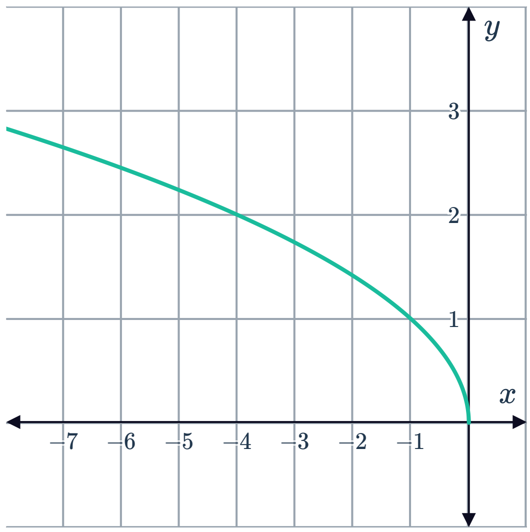
d Sketch the graph.

Domain and range

8 Consider the function $y = \sqrt{-x}$.

a State the domain of the function.

b State the range of the function.



9 Consider the function $y = \sqrt{x}$.

a Complete the table of values. Round any values to two decimal places if necessary.

x	0	1	2	3	4	5	9
y							

b State the domain of the function.

c State the range of the function.

d As x gets larger and larger, what value does y approach?

e Sketch the graph of $y = \sqrt{x}$.

10 Consider the function $y=-\sqrt{x}$.

a State the domain of the function.

b State the range of the function.

11 The function $y=\sqrt{x}$ has domain $x \geq 0$ and range $y \geq 0$.
What is the domain and range of $y=\sqrt{x}-2$?

12 Consider the function $y=\sqrt{x-5}$.

a State the domain of the function.

b State the range of the function.

c Do the functions $y=\sqrt{x}$ and $y=\sqrt{x-5}$ increase at the same rate?

13 Consider the function $y=\sqrt{-x}+6$.

a What is the smallest possible function value?

b State the domain of the function.

c State the range of the function.

14 A square root function has a range of $y \leq 0$ and a domain of $x \geq 0$. Determine whether the following could be the equation of the function:

a $y=\sqrt{x}$

b $y=-\sqrt{-x}$

c $y=-\sqrt{x}$

d $y=5\sqrt{x}$

e $y=-5\sqrt{x}$

f $y=\sqrt{-x}$

15 For each of the following functions:

i Sketch the graph

ii State the domain

iii State the range

a $f\left(x\right)=-\sqrt{x+1}$

b $f\left(x\right)=-2\sqrt{x+5}$

c $f\left(x\right)=3\sqrt{\left(\dfrac{1}{3}x\right)}$

d $f\left(x\right)=-\dfrac{\sqrt{x}}{2}-2$

e $f\left(x\right)=\dfrac{\sqrt{x-1}}{2}+2$

16 For each of the following functions:

i State the domain of the function.

ii State the range of the function.

iii Sketch the graph.

a $y=-\sqrt{x}+5$

b $y=-5\sqrt{x}$

c $y=-\sqrt{-x}$

d $y=\sqrt{x-3}+2$

17 For which values of x do the following expressions evaluate to a real number?

a $\sqrt{7x}$

b $\sqrt{x-2}$

c $\sqrt{3-x}$

d $\sqrt{15-5x}$

e $\sqrt{x^2+6}$

18 Consider the function $f(x)=\sqrt{x-2}+3$. State the domain of the function using interval notation.

Transformations

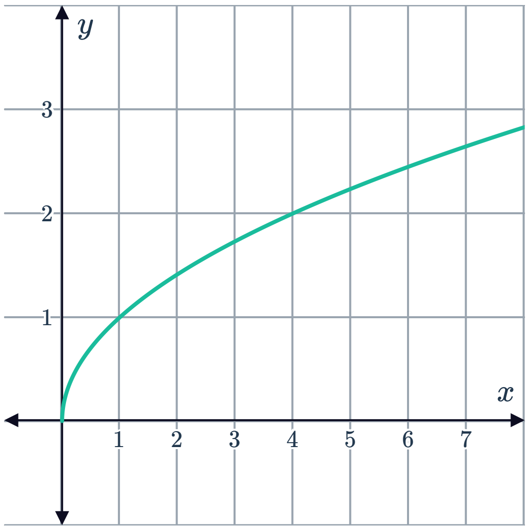
19 The graph of $y=\sqrt{x}$ has a vertex at $(0,0)$. By considering the transformations that have taken place, state the coordinates of the vertex of $y=-\sqrt{x}+3$.

20 The graph of $y=\sqrt{x}$ has been translated to the graph of $y=\sqrt{x}-4$.

a Describe the transformation that has occured on the original function.

b Hence, sketch the graph of $y=\sqrt{x}-4$.

21 Consider the graph of $y=\sqrt{x}$ shown:



a

i Sketch the curve after $y=\sqrt{x}$ has been reflected about the y-axis.

ii What is the equation of this new graph?

b

i Sketch the curve after $y=\sqrt{x}$ has been reflected about the x-axis.

ii What is the equation of this new graph?

22 Consider the function $y=\sqrt{x}$:

a Describe how we can transform the graph of $y=\sqrt{x}$ to get the graph of $y=\sqrt{x-4}+3$.

b Hence, sketch the graph of $y=\sqrt{x-4}+3$.

23 Sketch the curve $y=3\sqrt{x-2}+3$.